

VHDL:

the description of a complex system in the form of a diagram is illegible, quick changes to the system description

Languages: VHDL, Verilog, Abel

Advantages: more information than in a schematic description, shorter design cycle for complex systems, independence from technology, standardization, abstraction, simulation

Disadvantages: change of design style, costs of tools, training, synthesizability, problem, quality of synthesis, difficulty in assessing the complexity of the project

Levels of abstraction (style of describing systems)

Behavioral (algorithmic, systemic) - description of the the system's operation without implementation details (process, maps)

RTL (inter-register transfers, Dataflow)

logical (boolean, gate)

Structural (division into subsystems, multiple use of subsystems)

2 types of VHDL code

Concurrent code

Sequential code (process)

Syntax of vhdl:

Sample <= "0000"

next_sample <= sample

relational operators: =, /=, >, <, <=, >=

Signal A <= '0', '1' after 10 ns

$\underbrace{(\text{!} =)}_{\text{C}} = \underbrace{(\text{/} =)}_{\text{VHDL}}$

! = } assign a value to a variable

concatenation (gluing) operator: &

General rules:

VHDL is case-insensitive

A semicolon terminates an expression (the program line ending)

Each comment line begins with a letter and contains alphanumeric characters or an underscore (" _ ") in subsequent positions

The language is strongly type-oriented; automatic conversion is rare, overloading of operators, functions and procedures is possible.

Only a subset of the language is synthesizable.

Practise:

```
library IEEE;
```

```
use IEEE.std_logic_1164.all;
```

```
entity mux2to1 is
```

```
port (
```

```
    s      : in std_logic;
```

```
    x1, x0 : in std_logic_vector (7 downto 0);
```

```
    y      : out
```

```
);
```

```
end mux2to1
```

→ there is no semicolon

architecture of mux2to1 is

begin

$y \leftarrow x_1$ when ($S = '1'$)

 else x_0 ;

end mul ;